Firemap: A dynamic data-driven predictive **wildfire** modeling and visualization environment

D Crawl, J Block, K Lin, I Altintas - Procedia Computer Science, 2017 - Elsevier

… predict the **behavior** of the … **simulation**, **prediction**, and visualization of **wildfire behavior**. One goal of WIFIRE is to provide the tools to predict a more accurate rate of a spreading **wildfire**. …

☆ Save 99 Cite Cited by 43 Related articles

Adaptive **wildfire spread prediction** for complex terrain: modeling the effectiveness of sprinkler systems

J Kim, J Ahn, J Kang - Fire Ecology, 2024 - Springer

… of a comprehensive CFD **model** in mitigating **wildfire spread** using sprinkler systems as an … in **wildfire** within 10 h of ignition, significant influence of wind speed on **spread** patterns, and a …

☆ Save 99 Cite Cited by 7 Related articles All 8 versions

Understanding California **wildfire** evacuee **behavior** and joint choice making

SD Wong, JC Broader, JL Walker, SA Shaheen - Transportation, 2023 - Springer

… We first briefly **review** the literature on evacuation **behavior** … current literature available on **wildfire** evacuation **behavior**. … the **Wildfire** Decision **Model**, where people in a **wildfire** situation …

☆ Save 99 Cite Cited by 75 Related articles All 15 versions

Global **wildland fire management** research needs

PF Moore - Current Forestry Reports, 2019 - Springer

… fireline intensities from 20,000 to 60,000 kW/m and a rate of **spread** of 65 m/min (~ 4 km/h) [28•… The focus of this **review** is on global **wildland fire management** research needs for IFM; the …

☆ Save 99 Cite Cited by 62 Related articles All 3 versions

Analysis of evacuation **behavior** in a **wildfire** event

T Toledo, I Marom, E Grimberg, S Bekhor - International journal of disaster …, 2018 - Elsevier

… place, is critical for the efficient **management** of the event. The usual … A large-scale **wildfire** took place in Haifa on November 24, … Shortly after the fire events, a web **survey** was developed …

☆ Save 99 Cite Cited by 143 Related articles All 5 versions

Role of flow topology in wind-driven **wildfire propagation**

S Viknesh, A Tohidi, F Afghah, R Stoll, A Arzani - Physics of Fluids, 2025 - pubs.aip.org

… In a comprehensive **review**, Sullivan (2009a) detailed the development of physics-based … Gaining such fundamental insights is crucial, as it enables the **prediction** of **wildfire spread** …

☆ Save 99 Cite Cited by 1 Related articles All 4 versions

Prediction of **wildfire** rate of **spread** in grasslands using machine learning methods

S Khanmohammadi, M Arashpour… - … Modelling & Software, 2022 - Elsevier

… **Prediction** of **wildfire propagation** plays a crucial role in reducing … models to predict the rate of **spread** of grassfires. A dataset from … These models produced excellent fits to the **model** …

☆ Save 99 Cite Cited by 44 Related articles All 6 versions

LANDFIRE: a nationally consistent vegetation, **wildland fire**, and fuel assessment

MG Rollins - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… of different **wildfire management** strategies, **wildfire** season … ; (3) communications and iterative **review** by local ecologists and … the probability of individual **wildfire spread** and severity and …

☆ Save 99 Cite Cited by 788 Related articles All 14 versions

A simple physical **model** for **forest fire spread**

E Koo, P Pagni, J Woycheese… - In: Gottuk, D.; …, 2005 - research.fs.usda.gov

… mechanisms of **forest fire spread** and introduced a simple physics-based fire **spread model** for … Pitts [12] provided a thorough **review** of wind effects on fire. Weise and Biging [13] built a …

☆ Save 99 Cite Cited by 97 Related articles All 10 versions »

Improving **wildland fire spread prediction** using deep U-Nets

F Khennou, MA Akhloufi - Science of Remote Sensing, 2023 - Elsevier

… of 97.7%, the **model** surpassed the literature by … **model**. The proposed approach was implemented using consecutive forest **wildfire** perimeters, satellite images, Digital Elevation **Model** …

☆ Save 99 Cite Cited by 20 Related articles All 3 versions